LAGUNA STATE POLYTECHNIC UNIVERSITY

Bachelor of Science in Information Technology

BIOS vs. UEFI

Comparison and Explanation

A concise technical comparison of legacy BIOS firmware and the Unified Extensible Firmware Interface.

Project Information	Details
Course	ITEP 414 - System Administration and Maintenance
Scope	Firmware, boot process, disk partitioning, security, speed, advantages and limitations
Prepared by	[Your Full Name]
Date	[Submission Date]

Prepared for ITEP 414 - System Administration and Maintenance

Working template: replace bracketed fields and add genuine lab evidence before submission.

1. Comparison Table

Aspect	Legacy BIOS	UEFI
Definition	Older firmware interface that initializes hardware and starts boot code using a legacy boot model.	Modern firmware interface that provides standardized boot and runtime services and can launch EFI applications.
Boot process	Typically performs POST, selects a boot device, and executes boot code from the disk's MBR.	Uses firmware boot entries and loads an EFI application, commonly from the EFI System Partition.
Maximum disk support	Classic MBR layouts are commonly limited to about 2 TiB when using 512-byte logical sectors.	Often paired with GPT, which supports disks larger than 2 TiB; practical limits depend on implementation and OS.
Partition style	Commonly associated with MBR and its primary-partition limits.	Commonly paired with GPT; UEFI firmware and operating-system support determine the exact supported combinations.
Security	No standardized Secure Boot mechanism in the legacy BIOS model.	Can support Secure Boot, which checks signatures of boot components when enabled and correctly configured.
Speed	Startup speed depends on firmware, devices, and boot configuration; legacy initialization may be less flexible.	May initialize and boot more efficiently on modern platforms, but actual speed depends on hardware and configuration.
Advantages	Broad compatibility with older hardware and legacy operating systems; simple boot model.	Modern boot architecture, GPT support, boot entries, extensibility, and optional Secure Boot.
Disadvantages	Legacy limitations, MBR capacity/partition constraints, and fewer standardized pre-boot security capabilities.	More complex firmware settings; compatibility issues can arise with old operating systems or incorrectly configured boot modes.
Modern usage	Mostly older systems and legacy compatibility scenarios.	Standard choice for most modern computers and server platforms.

2. Why UEFI Has Largely Replaced BIOS

UEFI has largely replaced legacy BIOS because modern computers need a more flexible and extensible firmware environment. BIOS remains important for understanding older systems, but its traditional boot model is closely associated with the Master Boot Record (MBR), which has practical limitations for large disks and partition organization. With the common combination of UEFI firmware and the GUID Partition Table (GPT), systems can boot from disks larger than 2 TiB and use a more flexible partition layout.

UEFI also defines a standardized interface between firmware and operating-system boot software. Instead of relying only on a small boot-code area in the MBR, UEFI can use firmware boot entries and load EFI applications from an EFI System Partition. This design supports more structured boot management and is better suited to modern operating systems, storage devices, and server platforms.

Security is another major reason for UEFI adoption. Secure Boot is a UEFI-based capability that can verify signatures of boot components and prevent untrusted boot software from starting when it is enabled and correctly configured. Secure Boot is not automatic proof that a whole computer is secure: its protection depends on firmware configuration, trusted keys, operating-system support, and the rest of the security design.

UEFI may also offer faster or more configurable startup on some systems, but there is no universal speed guarantee. Actual boot time depends on firmware implementation, device discovery, hardware, boot configuration, and operating-system services. UEFI's most important benefits are its modern architecture, GPT compatibility, extensibility, and support for security features rather than speed alone.

In conclusion, UEFI is the preferred firmware model for modern computers and enterprise servers. BIOS knowledge remains useful for troubleshooting older equipment and understanding legacy MBR booting. Administrators should document the selected firmware mode because switching between BIOS and UEFI after installation can make an operating system unbootable if the partition layout and bootloader are not configured for the new mode.

3. Practical Notes for the Ubuntu Lab

- Record whether the VM uses BIOS or UEFI; do not assume the setting.
- Record the actual partition table and filesystems created by the installer.
- Do not switch firmware mode after installation unless you understand the bootloader and disk-layout implications.
- Use Secure Boot only when supported by the VM configuration and required by the lab; document whether it is enabled.

4. References

[1] UEFI Forum specifications. https://uefi.org/specifications. Accessed 26 October 2026.

[2] Microsoft Secure Boot overview. https://learn.microsoft.com/en-us/windows-hardware/drivers/bringup/secure-boot. Accessed 26 October 2026.

[3] Microsoft Secure Boot and Trusted Boot. https://learn.microsoft.com/en-us/windows/security/trusted-boot. Accessed 26 October 2026.

[4] Ubuntu Server installation documentation. https://ubuntu.com/server/docs/installation/. Accessed 26 October 2026.